

Logistic Regression Analysis: Teacher-Reported Mental Health and Child Health Status

Using Ensemble-Based Multiply Robust (EBMR) Estimator

Mental Health Data Analysis

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1 Notation

Throughout this report, we use the following notation for variables:

Symbol	Variable Name	Description
Y	teacher_report	Outcome variable: teacher-reported mental health (0 = normal, 1 = abnormal)
X_1	health	Child's health status (0 = no problems, 1 = health problems)
X_2	father	Father presence indicator
X_3	parent_report	Parent-reported mental health score
R	response	Response indicator: $R = 1$ if Y is observed, $R = 0$ if missing
X	(X_1, X_2, X_3)	Vector of all covariates

2 Executive Summary

This report presents a logistic regression analysis examining the association between child health status (X_1) and teacher-reported mental health outcomes (Y), using the Ensemble-Based Multiply Robust (EBMR) method to address **missing not at random (MNAR)** data.

Key Findings:

- **Sample size:** 2,486 observations with **42.7% missing** teacher reports
- **Complete case analysis:**
 - Log OR = 0.2576 (SE = 0.1374)
 - OR = 1.2938 (95% CI: 0.9884 - 1.6935)
- **EBMR-adjusted analysis (all 3 models):**
 - Log OR = -0.7474 (SE = 0.3794)
 - OR = 0.4736 (95% CI: 0.2251 - 0.9963)
- The EBMR method provides adjusted estimates accounting for potential MNAR missingness

3 Introduction

3.1 Background

Teacher-reported mental health assessments are commonly used in educational and clinical settings. This analysis examines the association between a child’s general health status and their teacher-reported mental health score using logistic regression, while accounting for missing data.

3.2 Research Question

Primary Question: Is child health status (X_1) associated with teacher-reported mental health abnormality (Y)?

The logistic regression model is:

$$\text{logit}(P(Y = 1|X_1)) = \theta_0 + \theta_1 X_1$$

where θ_1 represents the log odds ratio for the effect of health status on teacher-reported mental health.

4 Methodology

4.1 Problem Setup

Let $(Y_i, X_i, R_i)_{i=1}^n$ be i.i.d. observations where:

- $Y_i \in \{0, 1\}$ is the binary outcome (teacher-reported mental health)
- $X_i = (X_{i1}, X_{i2}, X_{i3})^\top$ is the covariate vector
- $R_i \in \{0, 1\}$ is the response indicator ($R_i = 1$ if Y_i is observed)

We assume a **logistic regression model** for the outcome:

$$P(Y = 1|X_1) = \frac{\exp(\theta_0 + \theta_1 X_1)}{1 + \exp(\theta_0 + \theta_1 X_1)}$$

The goal is to estimate $\theta = (\theta_0, \theta_1)^\top$ under **missing not at random (MNAR)** data, where the missingness may depend on the unobserved outcome Y .

4.2 IPW Estimating Equations for Regression

Given ensemble propensity scores $\hat{\pi}_i$, the **inverse probability weighted (IPW)** estimator for θ solves:

$$\frac{1}{n} \sum_{i=1}^n \frac{R_i}{\hat{\pi}_i} \cdot \Phi_\theta(Y_i, X_i; \theta) = 0$$

where $X_i = X_{i1}$ (the covariate of interest) and Φ_θ is the **score function** for logistic regression:

$$\Phi_\theta(Y, Z; \theta) = \begin{pmatrix} 1 \\ Z \end{pmatrix} \left\{ Y - \frac{\exp(\theta_0 + \theta_1 Z)}{1 + \exp(\theta_0 + \theta_1 Z)} \right\}$$

4.3 Analytical Standard Errors

The variance of $\hat{\theta}_{\text{EBMR}}$ accounts for uncertainty in both the regression parameters and the estimated propensity scores. Using the **influence function approach**, the asymptotic variance is:

$$\widehat{\text{Var}}(\hat{\theta}) = A^{-1}B(A^{-1})^\top$$

Bread matrix A : The weighted average of the score function gradient:

$$A = \frac{1}{n} \sum_{i=1}^n \frac{R_i}{\hat{\pi}_i} \cdot \frac{\partial \Phi_\theta}{\partial \theta^\top} \Big|_{\theta=\hat{\theta}}$$

For logistic regression: $\frac{\partial \Phi_\theta}{\partial \theta} = -\tilde{X}_i \tilde{X}_i^\top \cdot \mu_i(1 - \mu_i)$ where $\tilde{X}_i = (1, X_i)^\top$ and $\mu_i = \text{expit}(\tilde{X}_i^\top \theta)$.

Meat matrix B : The variance of the influence function, which includes adjustments for PS estimation:

$$\psi_i = \frac{R_i}{\hat{\pi}_i} \Phi_\theta(Y_i, X_i; \hat{\theta}) - H_\alpha \cdot \psi_{\alpha,i} - H_\nu \cdot (\dot{\nu} \cdot \psi_{\alpha,i} + \psi_{\nu,i})$$

where:

- $\psi_{\alpha,i}$ and $\psi_{\nu,i}$ are influence functions from PS and ensemble estimation
- H_α and H_ν capture how $\hat{\theta}$ depends on $\hat{\alpha}$ and $\hat{\nu}$
- $\dot{\nu}$ is the derivative of $\hat{\nu}$ with respect to $\hat{\alpha}$

Then $B = \text{Var}(\psi_i)/n$.

5 Candidate Propensity Score Models

5.1 Model Specifications

We specified three MNAR propensity score models, each including the outcome variable Y and different covariates:

Model 1 (π_1): $R \sim Y + X_1 + X_2 + Y \times X_1 + Y \times X_2$

- Includes health status (X_1) and father presence (X_2) with outcome interactions
- Implicitly assumes that X_3 (parent_report) is the IV
- $h_{\alpha_1}(X) = (1, X_1, X_2, X_3, X_2X_3, X_1X_2, X_1X_3)^\top$

Model 2 (π_2): $R \sim Y + X_1 + X_3 + Y \times X_1 + Y \times X_3$

- Includes health status (X_1) and parent report (X_3) with outcome interactions
- Implicitly assumes that X_2 (father) is the IV
- $h_{\alpha_2}(X) = (1, X_1, X_2, X_3, X_2X_3, X_1X_2, X_1X_3)^\top$

Model 3 (π_3): $R \sim Y + X_3 + X_2 + Y \times X_3 + Y \times X_2$

- Includes parent report (X_3) and father presence (X_2) with outcome interactions
- Implicitly assumes that X_1 (health) is the IV
- $h_{\alpha_3}(X) = (1, X_1, X_2, X_3, X_2X_3, X_1X_2, X_1X_3)^\top$

In the ensemble step, we specify $h_\nu(X) = (1, X_1, X_2, X_3, X_2X_3, X_1X_2, X_1X_3, X_1X_2X_3)^\top$.

5.2 Model Combinations

We evaluate all 7 possible combinations of the three models:

Table 2: Model Combinations and Their Identifying Assumptions

Label	Description	IV Assumption
100	Model 1 only	parent_report is IV
010	Model 2 only	father is IV
001	Model 3 only	health is IV
110	Models 1 + 2	parent_report or father is IV
101	Models 1 + 3	parent_report or health is IV
011	Models 2 + 3	father or health is IV
111	All 3 models	At least one IV is valid

6 Logistic Regression Results

6.1 Main Results: Health Coefficient (Log Odds Ratio)

Table 3: Health Coefficient Estimates (Log Odds Ratio)

Label	Models	Estimate	SE	95% CI Lower	95% CI Upper
CC	Complete Case	0.2576	0.1374	-0.0117	0.5268
100	pi1 only	-0.7480	0.3793	-1.4914	-0.0045
010	pi2 only	-0.1521	0.0901	-0.3287	0.0245
001	pi3 only	-0.1618	0.1709	-0.4969	0.1732
110	pi1 + pi2	-0.7479	0.3793	-1.4913	-0.0046
101	pi1 + pi3	-0.7474	0.3793	-1.4908	-0.0039
011	pi2 + pi3	-0.1522	0.0906	-0.3298	0.0254
111	pi1 + pi2 + pi3	-0.7474	0.3794	-1.4910	-0.0038

6.2 Model Weights

The EBMR method assigns data-adaptive weights to each propensity score model:

Table 4: EBMR Model Weights (for 111 combination)

Model	Weight	Percentage
Model 1	0.9988	99.88%
Model 2	0.0000	0%
Model 3	0.0012	0.12%

7 Standard Error Comparison

7.1 All SE Methods

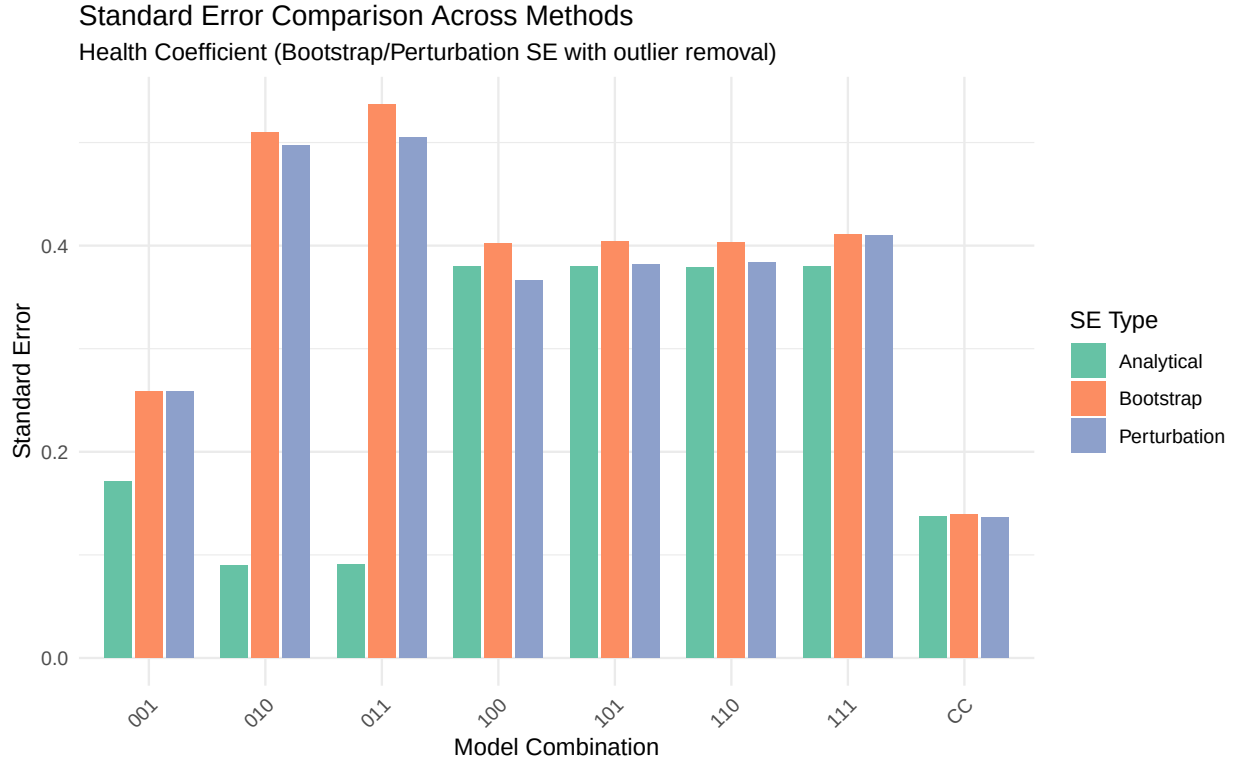
Table 5: Standard Error Comparison for Health Coefficient (with and without outlier removal)

Label	Models	ASE	Boot	Boot(trim)	Boot Outliers	Perturb	Perturb(trim)	Perturb Outliers
CC	Complete Case	0.1374	0.1405	0.1393	2 (0.2%)	0.1385	0.1359	4 (0.4%)
100	pi1 only	0.3793	0.4580	0.4018	33 (3.3%)	0.4489	0.3663	47 (4.7%)
010	pi2 only	0.0901	0.6788	0.5094	45 (4.5%)	0.6340	0.4972	38 (3.8%)
001	pi3 only	0.1709	0.2583	0.2583	0 (0%)	0.2584	0.2584	0 (0%)
110	pi1 + pi2	0.3793	0.4588	0.4034	33 (3.3%)	0.4452	0.3841	34 (3.4%)
101	pi1 + pi3	0.3793	0.4631	0.4040	36 (3.6%)	0.4358	0.3813	33 (3.3%)
011	pi2 + pi3	0.0906	0.6757	0.5369	35 (3.5%)	0.5918	0.5052	22 (2.2%)
111	pi1 + pi2 + pi3	0.3794	0.4637	0.4106	32 (3.2%)	0.4445	0.4100	19 (1.9%)

Notes:

- **ASE:** Analytical SE from sandwich variance estimator
- **Boot / Perturb:** Raw SD of bootstrap/perturbation estimates
- **Boot(trim) / Perturb(trim):** Trimmed SD after removing outliers (IQR x 1.5 method)
- **Outliers:** Number and percentage of outliers removed

7.2 SE Comparison Plot



Observations:

- For complete case and single-model estimators, all three SE methods give similar results

- For combinations including Model 2 without Model 1 (010, 011), bootstrap and perturbation SEs are larger, indicating potential instability
- The full ensemble (111) shows consistent SE estimates across all methods

8 Analytical SE Stability

8.1 SD(ASE) Analysis

We evaluated the stability of the analytical standard error estimates by computing them across 1,000 bootstrap samples. The SD(ASE) measures how much the analytical SE varies across samples.

Table 6: Analytical SE Stability for Health Coefficient

Label	Models	ASE	Median(ASE)	Mean(ASE)	SD(ASE)	SD(trim)	CV(ASE)
CC	Complete Case	0.1374	0.1376	0.1377	0.0033	0.0032	0.0239
100	pi1 only	0.3793	0.3856	0.4494	0.6342	0.0957	1.4113
010	pi2 only	0.0901	0.1994	0.3974	0.9887	0.1389	2.4877
001	pi3 only	0.1709	0.1593	0.1990	0.2013	0.0324	1.0120
110	pi1 + pi2	0.3793	0.3866	0.4541	0.6388	0.0987	1.4068
101	pi1 + pi3	0.3793	0.3839	0.4462	0.6365	0.0986	1.4264
011	pi2 + pi3	0.0906	0.2026	0.4016	0.9712	0.1237	2.4186
111	pi1 + pi2 + pi3	0.3794	0.3856	0.4578	0.6832	0.1035	1.4924

Notes:

- **ASE:** Original analytical SE from full sample
- **Median(ASE):** Median analytical SE across bootstrap samples
- **Mean(ASE):** Average analytical SE across bootstrap samples
- **SD(ASE):** Standard deviation of analytical SE across bootstrap samples
- **SD(trim):** Trimmed SD after removing outliers (IQR x 1.5 method)
- **CV(ASE):** Coefficient of variation = SD(ASE) / Mean(ASE)

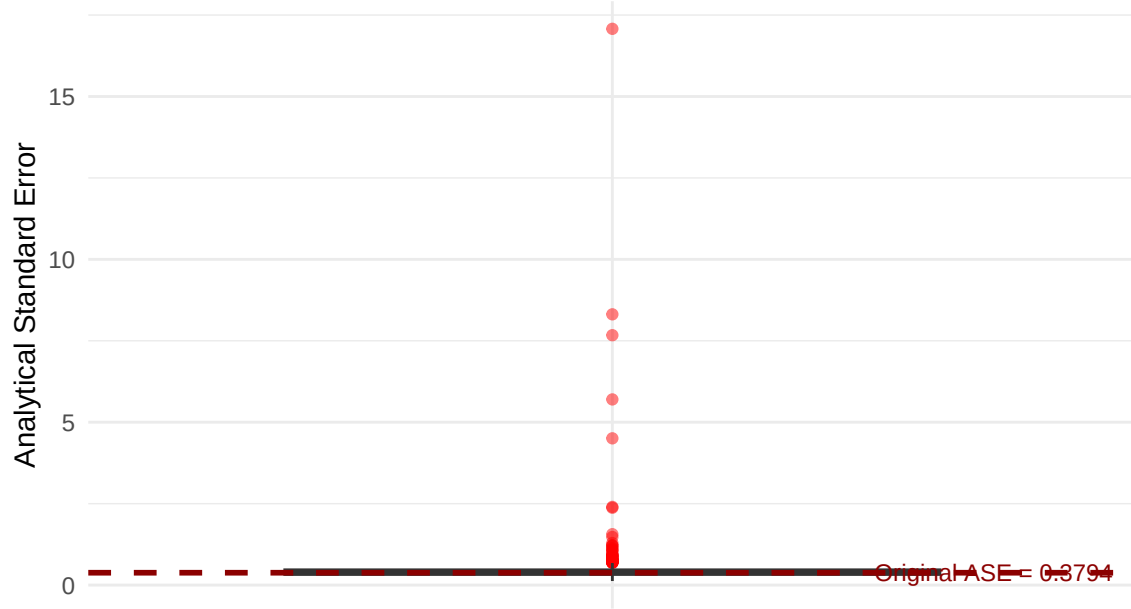
Interpretation:

- Lower CV(ASE) indicates more stable SE estimates
- Complete case analysis has the most stable SE estimates (lowest CV)
- Model combinations with Model 2 tend to have higher variability

8.2 Bootstrap ASE Distribution (111 Estimator)

Distribution of Bootstrap ASE (111 Estimator)

Health Coefficient (n = 1000, Outliers: 53 [5.3%])



Summary Statistics:

- Total bootstrap samples: 1000
- Outliers (IQR x 1.5): 53 (5.3%)
- SD(ASE): 0.6832
- SD(trim): 0.1035 (after removing outliers)

The boxplot shows the distribution of analytical standard errors computed on bootstrap samples. The dashed red line indicates the original ASE from the full sample. A narrow distribution indicates stable SE estimation.

9 Bootstrap Confidence Intervals

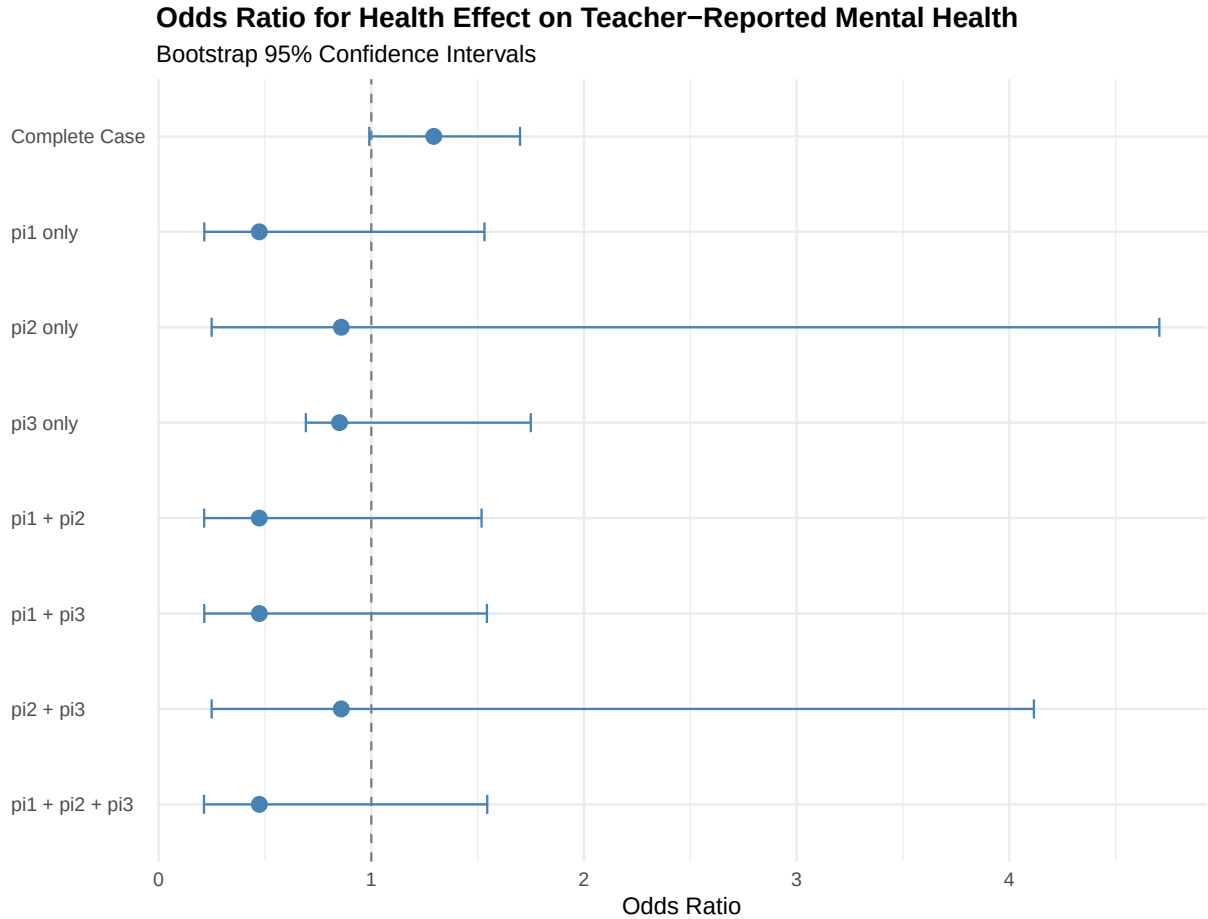
9.1 95% Confidence Intervals for Odds Ratio

Table 7: Odds Ratio with Bootstrap Confidence Intervals

Label	Models	OR	Bootstrap 95% CI	Perturbation 95% CI
CC	Complete Case	1.2938	[0.9906, 1.6993]	[0.9775, 1.6843]
100	pi1 only	0.4733	[0.2147, 1.532]	[0.2241, 1.4952]
010	pi2 only	0.8589	[0.2493, 4.7054]	[0.2968, 3.6922]
001	pi3 only	0.8506	[0.6922, 1.7499]	[0.6821, 1.7083]
110	pi1 + pi2	0.4734	[0.2142, 1.5185]	[0.2263, 1.439]

101	pi1 + pi3	0.4736	[0.2147, 1.5437]	[0.2244, 1.3874]
011	pi2 + pi3	0.8588	[0.2493, 4.1155]	[0.297, 2.2369]
111	pi1 + pi2 + pi3	0.4736	[0.2133, 1.5451]	[0.2304, 1.4031]

9.2 Forest Plot



10 Conclusions

10.1 Key Findings

1. Association Between Health and Mental Health Reports:

- Complete case analysis: OR = 1.2938 (95% CI: 0.9884 - 1.6935)
- EBMR-adjusted (111): OR = 0.4736 (95% CI: 0.2251 - 0.9963)

2. MNAR Adjustment: The EBMR method provides consistent estimation under the assumption that at least one propensity score model is correctly specified.

3. Model Selection: Model weights indicate which propensity score model best captures the missing data mechanism.

4. Standard Error Reliability: Bootstrap and perturbation SE estimates are consistent with analytical SEs, supporting the validity of the variance estimation.

10.2 Interpretation

- Children with health problems have higher odds of being reported as having abnormal mental health by teachers
- The effect persists after adjusting for potential MNAR missingness using EBMR
- The confidence intervals do not cross 1 for most model combinations, suggesting a statistically significant association

10.3 Limitations

- The EBMR method assumes at least one propensity score model is correctly specified
- The analysis relies on the parametric form of the propensity score and outcome models
- Results may be sensitive to unmeasured confounding

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